

Deterministic Optimization

Large-Scale Optimization

Cutting Stock Problem

Andy Sun

Assistant Professor

Stewart School of Industrial and Systems Engineering

Cutting Stock Problem

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Learning Objectives for this module

- Discover new applications of linear programming in large-scale optimization
- Explore the famous cutting-stock problem with large-scale LP formulation
- Discover column generation technique

Paper Company's Operational Problem

- A paper company produces large rolls of paper
- Each roll has width W
- These large rolls of paper need to be cut into smaller rolls to sell
- Demand from customers come in as orders of smaller roll widths

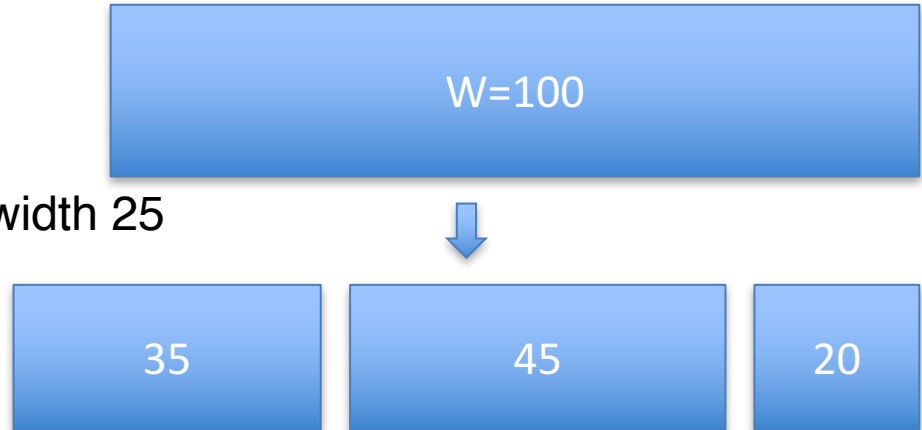


Demand of Smaller Rolls

- The big roll of paper has width $W = 100$
- The company receives orders from customers
- The orders are:
 - 150 rolls of width 25
 - 200 rolls of width 35
 - 300 rolls of width 45

Patterns to Cut Large Rolls

- In order to meet demand, paper company needs to cut large rolls into smaller ones
- For example:
 - Cut 1 large roll into 1 smaller roll of width 35, 1 smaller roll of width 45, 0 smaller roll of width 25
 - The remaining 20 is waste
 - Denote this way of cutting as
 - $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$



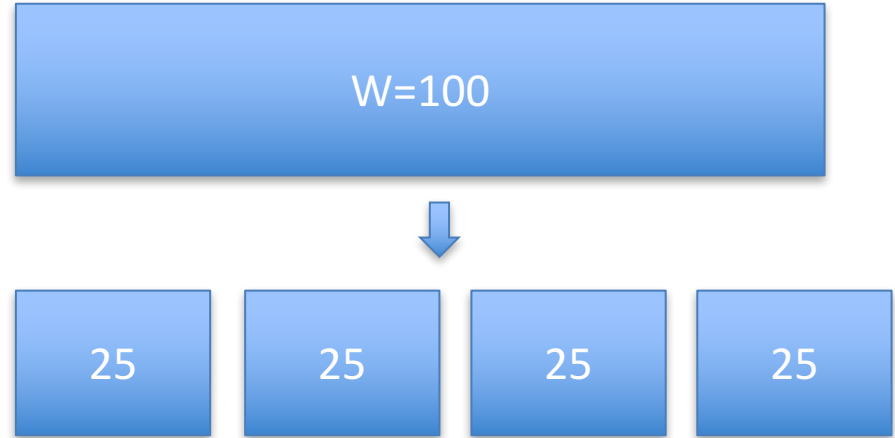
Patterns to Cut Large Rolls

- In order to meet demand, paper company needs to cut large rolls into smaller ones
- For example:
 - Cut 1 large roll into 2 smaller rolls of width 45,
 - The remaining 10 is waste
 - Denote this way of cutting as
 - $\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$



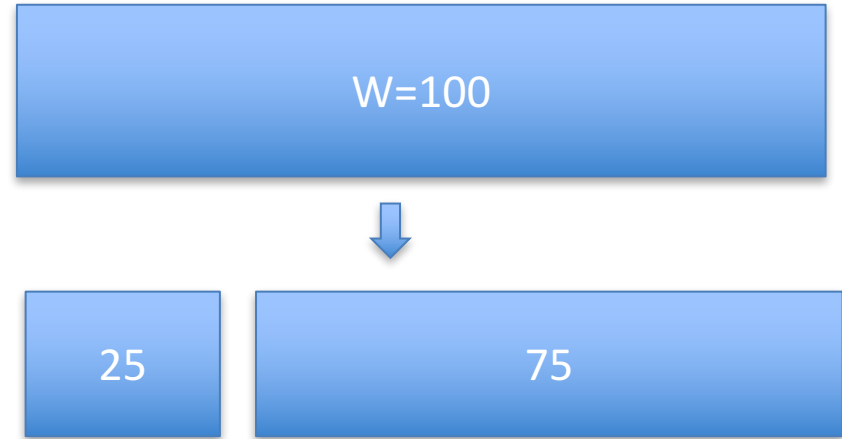
Patterns to Cut Large Rolls

- In order to meet demand, paper company needs to cut large rolls into smaller ones
- For example:
 - Cut 1 large roll into 4 smaller rolls of width 25,
 - No remaining paper
 - Denote this way of cutting as
 - $\begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$



Patterns to Cut Large Rolls

- In order to meet demand, paper company needs to cut large rolls into smaller ones
- For example:
 - Cut 1 large roll into 1 smaller rolls of width 25,
 - Remaining width 75 is waste
 - Denote this way of cutting as
 - $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
 - Not very economic, but a feasible way



Cutting Stock Problem

- How should the paper company cut the large rolls to satisfy demand in the most economic way?
- Different ways to measure economy:
 - Total number of large rolls
 - Amount of waste paper
 - Smallest number of switches between different patterns
- We choose to minimize the total number of large rolls used and will formulate an optimization problem in the next lecture.

Summary

- We learned:
 - Cutting stock problem
 - The meaning of a pattern